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OOPS Assignment 7

1. Try out all the code demonstrated in the lectures.

```
pigeon can fly
This is a bird
red
RED the color is: red
BLUE the color is: blue
GREEN the color is: green
```

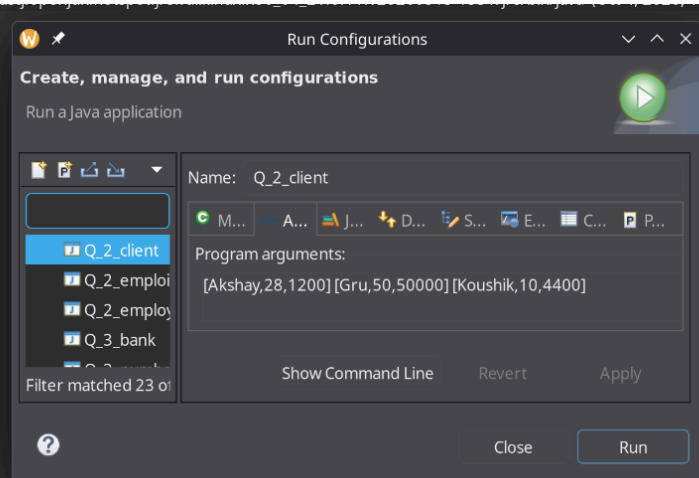
2. Create a class named Employee with an integer field id. You can also have other fields of your interest. The class should have proper constructors and get/set methods.
 - a. Override the finalize() method in the Employee class so that it prints: "Employee [id] is being garbage collected!".
 - b. Properly override the equals() method of the Object class in the Employee class. (1 point)
 - c. Have a client class and in the main method of it, create 3 employees.
 - i. Read the details of all the employees from the java command-line arguments. (1 point) 1. Note: Integer.parseInt() is a built-in static method used to convert a String into a primitive int data type.
 - ii. Have the necessary validations in the code. (1 point)
 - d. Demonstrate one of the employees being garbage collected. (1 point)

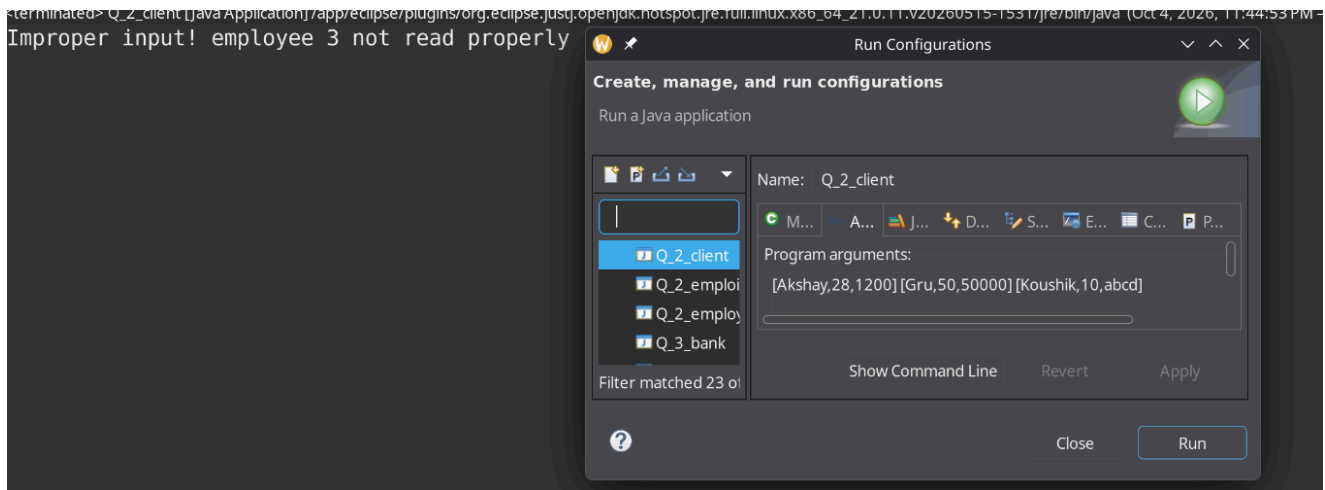
```
Employee ID: 28
Employee name: Akshay
Employee Salary: 1200.0

Employee ID: 50
Employee name: Gru
Employee Salary: 50000.0

Employee ID: 10
Employee name: Koushik
Employee Salary: 4400.0

false
Employee 10 is being garbage collected!
The program has ended
```





3. We are building a backend for a smart home system. You have an outer class representing a

SmartHouse and a tightly coupled inner class representing a SmartThermostat. (2 points) a. Let SmartHouse have houseName and currentTemperature variables. Have relevant constructors and get/set methods. b. Create the non-static inner class SmartThermostat. c. Write a method called adjustTemperature(int targetTemp) in the SmartThermostat class i. This method should directly modify the outer class's currentTemperature variable and print out a confirmation message such as: "[houseName] Thermostat: Changing temperature from X°C to Y°C." d. Demonstrate all the features in the client/Main class.

```
House Name: Batcave
Temperature: 0.0
Enter the temperature of room to be set
18
Batcave Thermostat: Changing temperature from 0.0 to 18.0
House Name: Batcave
Temperature: 18.0
```

4. You are tasked with building a control subsystem for an advanced, modern Smart Robot Device.

This device needs to handle distinct functionalities such as charging power, processing logic, and environmental clean-up.

a. First setup the interfaces. (1 point)

- i. Create a base interface named PowerDevice with an abstract method void turnOn() and a default void checkBattery() method that prints "Battery status: Good".
- ii. Create another base interface named SmartDevice with an abstract method void connectToWiFi().
- iii. Create a third interface named AmphibiousCleaner that extends both PowerDevice and SmartDevice. Add a new abstract method void deepClean().

- b. Create a concrete class named SmartRobotVacuum that implements both PowerDevice and SmartDevice directly. (1 point)
- i. Provide concrete implementations for turnOn() and connectToWiFi() that print distinct status actions.
- c. Have a client/main class and in the main method do the following. (2 points)
- i. Create an anonymous class that implements AmphibiousCleaner.
 1. Override the default checkBattery() method to provide custom behaviour.
 - ii. Instantiate the anonymous class of the AmphibiousCleaner interface on the fly without declaring a separate physical class file.
 1. Implement all required inherited abstract methods inside this anonymous block.
 - iii. Call all the methods (turnOn(), connectToWiFi(), deepClean(), and your overridden checkBattery()) to verify your architecture works.

```
The system is connected to wifi
The system is connected to internet
Performing Deep Clean
Battery at 80% capacity
```